Mark Stanley

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EDUCATION

University of Wisconsin-Madison May 2027
B.S. in Computer Science, Mathematics, and Philosophy *GPA: 3.76/4.00*
Related Courses: Deep Learning for NLP, Artificial Intelligence, Computational Learning Theory, Applied Linear Algebra, Probability Theory, Real Analysis, Algorithms, Big Data Systems, Language and Meaning, Graduate Seminar: Philosophy of Mind

EXPERIENCE

Undergraduate Computer Vision Research Assistant September 2025 - Present
Guided by Prof. Yong Jae Lee *Madison, WI*
- Co-authored *MuRF: Unlocking the Multi-Scale Potential of Vision Foundation Models*, introducing a training-free inference strategy to fuse multi-resolution features in frozen models like DINOv2.
- Designed and implemented dense-task evaluation pipelines, including depth estimation benchmarks on NYUv2 and KITTI, and semantic segmentation on Pascal VOC using PyTorch.

Software Development Intern September 2025 - Present
Nostomania *Madison, WI*
- Implemented a geometric rectification model leveraging DinoV2 features to align comic book covers, creating a standardized pipeline for visualization and downstream variant classification.
- Architected a SQL-backed scraping pipeline of PCGS sales for algorithmic price prediction.

Software Development Intern May 2025 - August 2025
C-Motive Technologies *Middleton, WI*
- Developed two deep learning models using Anomalib, integrated into a custom preprocessing pipeline to robustly identify microscopic defects on high-resolution motor part images.
- Designed a column-wise ensemble inference system training independent models per image column with aggregated predictions, achieving ~0.98 pixel AUROC and AUPRO.

Software Technical Lead May 2024 - December 2024
Wisconsin Messenger *Madison, WI*
- Led a 20-person engineering organization to deliver a digital newspaper platform, successfully scaling distribution to over 47,000 students via a CI/CD-integrated cloud architecture.
- Directed the development of a Python data pipeline extracting 8,000+ articles weekly from 130+ feeds, applying NLP techniques to semantically filter, cluster, and recommend content.

PROJECTS

Deep Learning for Ultimate Tic-Tac-Toe (U3T) | *Python, PyTorch, Machine Learning*
- Engineered a high-throughput state-space environment for U3T, training PyTorch agents to learn position evaluation heuristics that optimize search-based decision making.
- Restructured an 8-million-example dataset generated via Monte Carlo Tree Search (MCTS) into an optimized compressed binary format, reducing overhead for training.

COMPUTER SKILLS

- **Programming Languages:** Python, Java, Javascript, TypeScript, C, SQL, LaTeX
- **Development Tools:** Git, Docker, Linux, Hugging Face, MLFlow, Wandb, Flask, FastAPI
- **Libraries:** PyTorch, Lightning, Transformers, TransformerLens, vLLM, datasets, Anomalib, OpenCV, Albumentations**,** timm, NumPy, SciPy, pandas, Matplotlib, PyArrow